XAI Project: Exploratory Data Analysis & Feature Selection

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1 Exploratory Data Analysis & Preprocessing

1.1 Introduction

This report presents the exploratory data analysis (EDA) and preprocessing steps performed on the diabetes dataset. The goal is to understand the data distribution, identify patterns, and prepare the dataset for machine learning modeling.

1.2 Dataset Overview

The dataset contains health-related features such as BMI, Age, Blood Pressure, and lifestyle indicators. The target variable is `Diabetes_binary` where $0 = Non-diabetic$ and $1 = Diabetic$.

1.3 Target Distribution

The dataset exhibits an imbalance between the two classes.

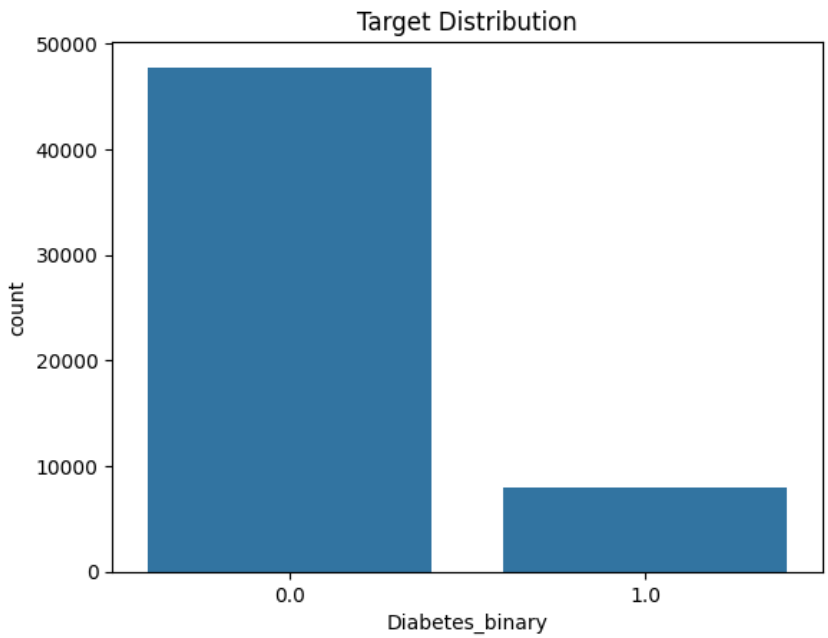

Figure 1: Target Distribution (Diabetes_binary)

1.4 Feature Distribution

The following distributions highlight the spread of key numerical and categorical features within the dataset.

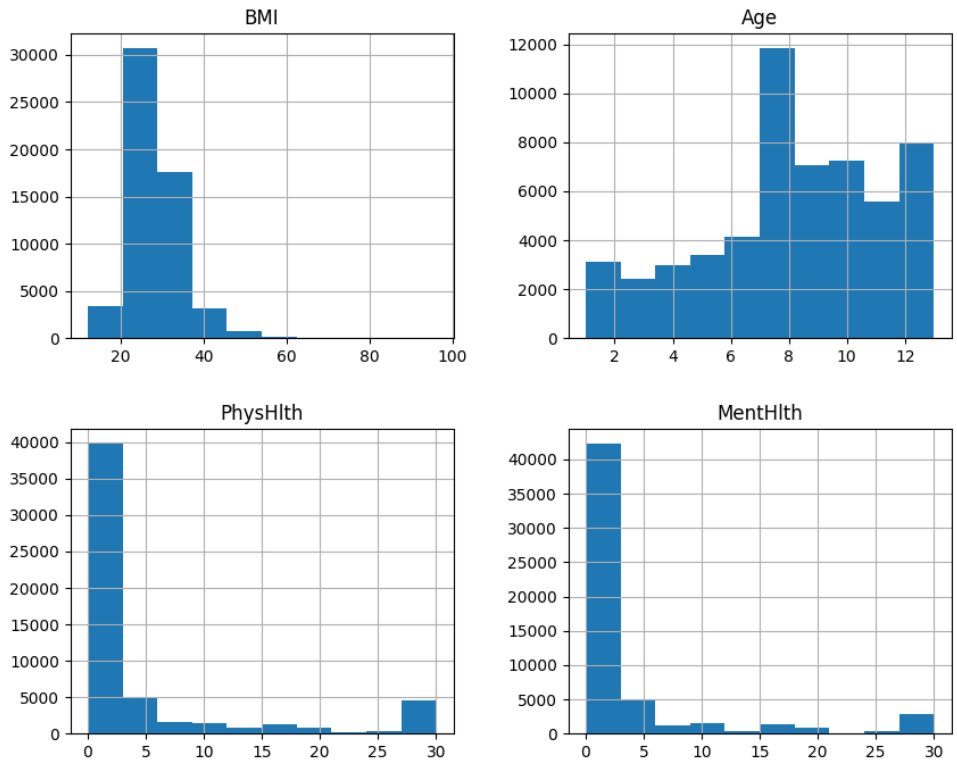

Figure 2: Distributions for BMI, Age, PhysHlth, and MentHlth

1.5 Feature vs Target

We analyze how specific features relate to the diabetes target variable.

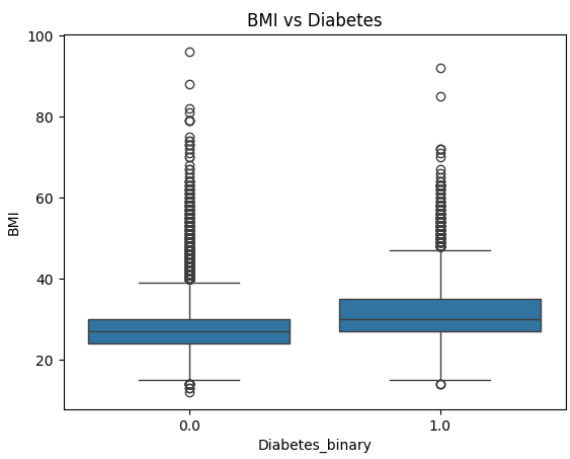

Figure 3: BMI vs Diabetes

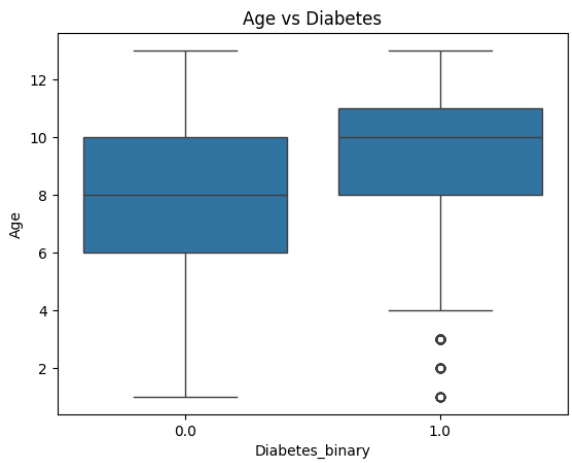

Figure 4: Age vs Diabetes

1.6 Correlation Analysis

- **Top correlated features:** GenHlth, HighBP, BMI.
- **Negative correlations:** Income, Education, Physical Activity.

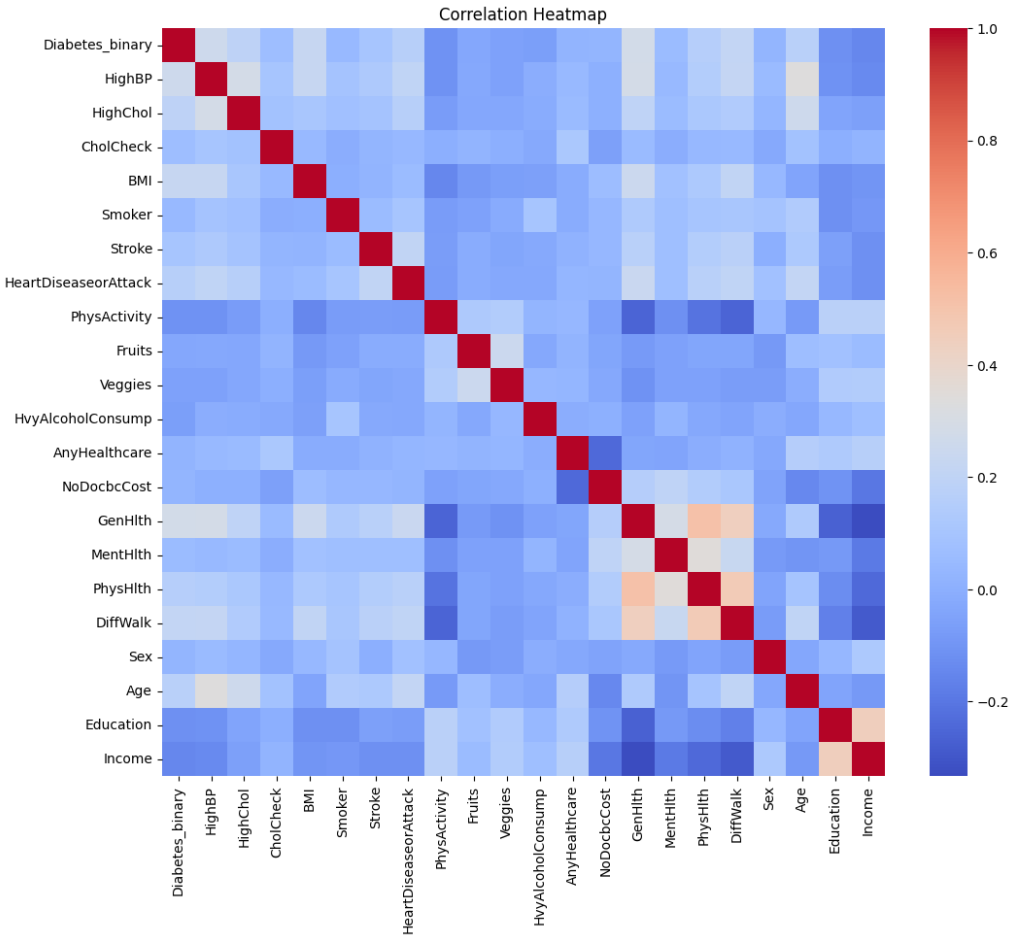

Figure 5: Correlation Heatmap of Features

1.7 Outliers Analysis

BMI contains outliers (which are valid medical cases), while `PhysHlth` and `MentHlth` are sparse.

1.8 Preprocessing

- Missing values handled
- Duplicates removed
- Scaling applied
- SMOTE used for imbalance

1.9 Challenges

- Data imbalance
- Skewed features
- Sparse variables

1.10 EDA Conclusion

Diabetes prediction depends on multiple factors; no single feature is sufficient, requiring advanced models.

2 Feature Selection

2.1 Overview

In this stage, feature selection was conducted prior to training any machine learning models. The main objective was to identify the most relevant features for predicting Type 2 Diabetes, reduce noise in the dataset, and improve both model performance and interpretability.

A filter-based approach was adopted, which included:

- Correlation analysis with the target variable
- Removal of redundant features
- Chi-square statistical feature selection

2.2 Methodology

2.2.1 Correlation Analysis with Target

Pearson correlation was computed between each feature and the target variable (`Diabetes_binary`) to assess the strength of their relationships.

Key Findings (Most correlated features):

- GenHlth (0.29)
- HighBP (0.26)
- BMI (0.21)
- Age (0.17)
- HeartDiseaseorAttack (0.17)

Interpretation: These results suggest that general health condition, blood pressure, obesity, and age are strongly associated with diabetes risk, which is consistent with established medical knowledge.

2.2.2 Redundancy Check

To address multicollinearity, correlations between the selected features were examined.

Result: No highly correlated feature pairs (correlation > 0.8) were identified.

Interpretation:

- The dataset does not contain redundant features.
- Each variable contributes distinct information.
- No features needed to be removed at this stage.

2.2.3 Chi-Square Feature Selection

The Chi-square test was applied to measure the statistical dependency between each feature and the target variable.

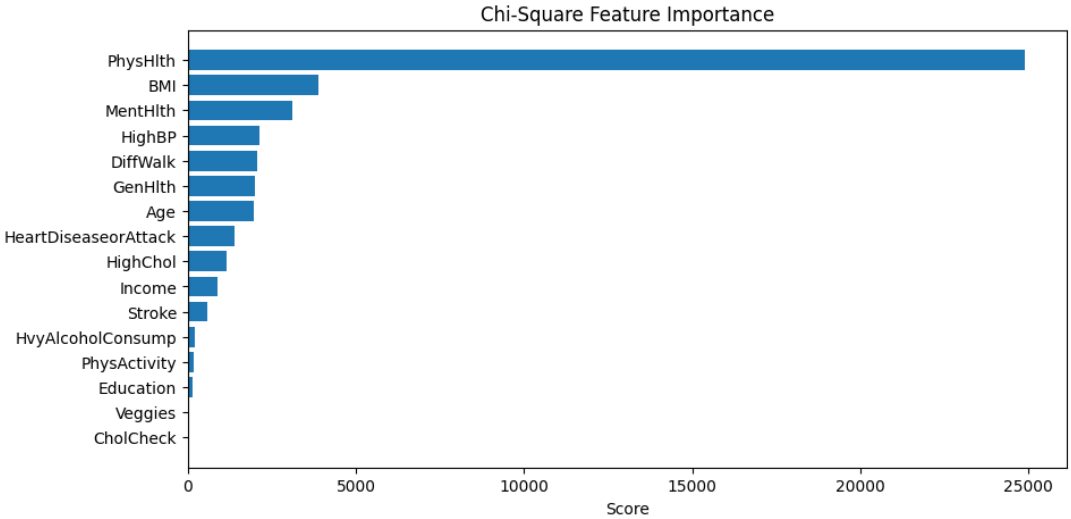

Figure 6: Chi-Square Feature Importance

Selected Features (Top 10): Based on the Chi-square test, the following features were selected: GenHlth, HighBP, DiffWalk, BMI, HighChol, Age, HeartDiseaseorAttack, PhysHlth, MentHlth, and Income.

2.3 Results Interpretation

The feature selection results show that several variables have a strong relationship with Type 2 Diabetes. The most important features based on the Chi-square test are:

- **PhysHlth and MentHlth**, which achieved the highest scores, indicating that physical and mental health conditions are strong predictors of diabetes.
- **BMI, HighBP, and HighChol**, which are well-known medical risk factors related to obesity and cardiovascular diseases.
- **Age and HeartDiseaseorAttack**, which reflect the impact of aging and existing health conditions on diabetes risk.

On the other hand, features such as CholCheck, Veggies, and Smoker showed relatively low scores, indicating a weaker contribution to prediction. Overall, the selected features are both statistically significant and medically meaningful, making them suitable for training machine learning models.

2.4 Conclusion

The feature selection process successfully identified the most informative variables for predicting Type 2 Diabetes. The selected features are both statistically significant and medically meaningful, ensuring that the dataset is well-prepared for model training.

This step contributes to:

- Improved model efficiency
- Reduced computational complexity
- Enhanced interpretability of results